

# Calculus II

## Weierstrass substitution theory

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# Integrals of the form $\int R(\cos \theta, \sin \theta) d\theta$ , $R$

Let  $R$  be an arbitrary rational function in two variables (quotient of polynomials in two variables).

## Question

*Can we integrate  $\int R(\cos \theta, \sin \theta) d\theta$ ?*

- Yes. We will learn how in what follows.
- The algorithm for integration is roughly:
  - Apply the substitution  $\theta = 2 \arctan t$  to transform to integral of rational function.
  - Solve as previously studied.

# The rationalizing substitution $\theta = 2 \arctan t$

Let  $R$ - rational function in two variables.  $\int R(\cos \theta, \sin \theta) d\theta$  can be integrated via the substitution  $\theta = 2 \arctan t$ . How does this transform  $\sin \theta$ ,  $\cos \theta$ ? How does this transform  $d\theta$ ? How is  $t$  expressed via  $\theta$ ?

$$\begin{aligned}\sin \theta &= \sin(2 \arctan t) = \frac{2 \tan(\arctan t)}{1 + \tan^2(\arctan t)} = \frac{2t}{1 + t^2} \\ \cos \theta &= \cos(2 \arctan t) = \frac{1 - \tan^2(\arctan t)}{1 + \tan^2(\arctan t)} = \frac{1 - t^2}{1 + t^2}\end{aligned}$$

Recall the expression of  $\sin(2z)$ ,  $\cos(2z)$  via  $\tan z$ :

$$\begin{aligned}\sin(2z) &= 2 \sin z \cos z = \frac{2 \sin z \cos z \frac{1}{\cos^2 z}}{(\cos^2 z + \sin^2 z) \frac{1}{\cos^2 z}} = \frac{2 \tan z}{1 + \tan^2 z} \cdot \\ \cos(2z) &= \cos^2 z - \sin^2 z = \frac{(\cos^2 z - \sin^2 z) \frac{1}{\cos^2 z}}{(\cos^2 z + \sin^2 z) \frac{1}{\cos^2 z}} = \frac{1 - \tan^2 z}{1 + \tan^2 z} \cdot\end{aligned}$$

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## Theorem

*The substitution given above transforms  $\int R(\cos \theta, \sin \theta) d\theta$  to an integral of a rational function of  $t$ .*

The integral  $\int \sec \theta d\theta$  appears often in practice. A quicker solution will be shown later, but first we show the standard method.

### Example

$$\text{Set } \theta = 2 \arctan t, \cos \theta = \frac{1 - \tan^2(\frac{\theta}{2})}{1 + \tan^2(\frac{\theta}{2})} = \frac{1 - t^2}{1 + t^2}, d\theta = 2 \frac{1}{1 + t^2} dt.$$

$$\begin{aligned} \int \sec \theta d\theta &= \int \frac{1}{\cos \theta} d\theta = \int \frac{1}{\left(\frac{1-t^2}{1+t^2}\right)} \frac{2}{(1+t^2)} dt \\ &= \int \frac{2}{1-t^2} dt = \int \left( \frac{1}{1-t} + \frac{1}{1+t} \right) dt \quad \left| \text{part. fractions} \right. \\ &= -\ln |1-t| + \ln |1+t| + C \\ &= \ln \left| \frac{1+t}{1-t} \right| + C \\ &= \ln \left| \frac{1 + \tan(\frac{\theta}{2})}{1 - \tan(\frac{\theta}{2})} \right| + C \end{aligned}$$

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Set  $\theta = 2 \arctan t$ ,  $\cos \theta = \frac{1 - \tan^2(\frac{\theta}{2})}{1 + \tan^2(\frac{\theta}{2})} = \frac{1 - t^2}{1 + t^2}$ ,  $d\theta = 2 \frac{1}{1 + t^2} dt$ .

$$\int \sec \theta d\theta = \ln \left| \frac{1 + \tan(\frac{\theta}{2})}{1 - \tan(\frac{\theta}{2})} \right| + C$$

This is a perfectly good answer, however there's a simplification:

$$\begin{aligned} \tan \theta + \sec \theta &= \frac{\sin \theta + 1}{\cos \theta} = \frac{2 \sin(\frac{\theta}{2}) \cos(\frac{\theta}{2}) + \sin^2(\frac{\theta}{2}) + \cos^2(\frac{\theta}{2})}{\cos^2(\frac{\theta}{2}) - \sin^2(\frac{\theta}{2})} \\ &= \frac{(\sin(\frac{\theta}{2}) + \cos(\frac{\theta}{2}))^2}{(\cos(\frac{\theta}{2}) - \sin(\frac{\theta}{2}))(\cos(\frac{\theta}{2}) + \sin(\frac{\theta}{2}))} \\ &= \frac{\sin(\frac{\theta}{2}) + \cos(\frac{\theta}{2})}{\cos(\frac{\theta}{2}) - \sin(\frac{\theta}{2})} = \frac{1 + \tan(\frac{\theta}{2})}{1 - \tan(\frac{\theta}{2})} \end{aligned}$$

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